Chest X-Ray Generation with a Residual VAE

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Abstract

A residual VAE was trained for chest X-ray generation. Direct samples from the standard normal prior looked fragmented, but fitting a Gaussian mixture to the training latent means improved generation. The final test probability FID was 0.0210, pooled FID 45.8 and Inception Score 1.96.

1 Model and training

Both classes of the Chest X-Ray Pneumonia dataset were used. Removing 26 duplicates left 5,190 training images. The official validation split has only 16 images, so 624 training images formed the holdout and 4,566 remained for optimization. Final evaluation used the official 624 test images. The 64×64 grayscale inputs kept training affordable. EXIF orientation was corrected before bilinear resizing and scaling to $[0, 1]$. Direct resizing changed the original aspect ratios.

The model has about 10.2 million parameters and a 256-dimensional latent space. Four encoder stages use 32, 64, 128 and 256 channels, with residual blocks, GroupNorm, SiLU and stride-two downsampling. Two output layers predict the Gaussian mean μ and log variance s. Latent samples take the form $z = \mu + \exp(s/2) \odot \epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$. The decoder brings this code back to an image through nearest-neighbor upsampling, 3×3 convolutions and residual blocks, ending with a sigmoid output mean.

The objective is the negative ELBO with a Gaussian pixel likelihood, fixed $\sigma = .01$ and KL weight one:

$$\mathcal{L} = \mathbb{E}_{q_\phi(z|x)}[-\log p_\theta(x \mid z)] + D_{\mathrm{KL}}(q_\phi(z \mid x) \| \mathcal{N}(0, I)). \tag{1}$$

Pixel and latent terms are summed per image, then averaged over the batch. Training ran for 300 epochs with Adam and batch size 64. The learning rate followed a cosine schedule from 10^{-3} to 10^{-5}, with gradient norms clipped at 5. The A30 run used float32 and no data augmentation.

The main difficulty was random generation: draws from $\mathcal{N}(0, I)$ often contained fragmented textures despite reasonable reconstructions. I therefore fitted a five-component, full-covariance Gaussian mixture to the training posterior means and sampled from it at temperature .85. For the epoch-300 model, holdout pooled FID fell from 161 with normal-prior sampling to 35.2 with this sampler.

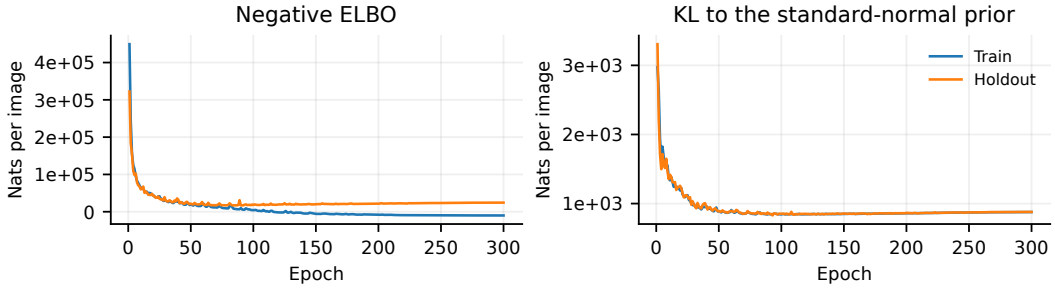


Figure 1: Training and holdout negative ELBO (left, linear scale) and KL (right).

2 Generated images and evaluation

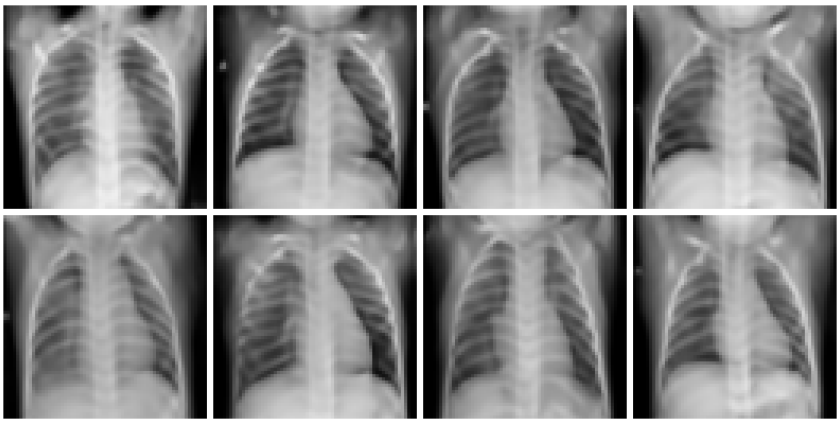


Figure 2: Eight selected samples from 5,000 generated images (seed 20261023).

Metric	Test result
Probability FID ↓	0.0210
Inception Score ↑	1.96
Pooled FID ↓	45.8

Table 1: Final test scores.

Epoch	FID	Train MSE	Holdout MSE
100	61.6	0.902	1.58
150	41.3	0.415	1.67
200	40.5	0.280	1.77
250	38.4	0.223	1.84
300	**35.2**	0.205	1.89

Table 2: Holdout FID and posterior-sampled MSE (x1,000).

Ignite evaluation used uint8 RGB images, bilinearly resized to 299×299. FID compares 624 test images with the first 624 generated images; IS uses all 5,000. The FID rows use Inception Softmax outputs and pooled features, respectively.

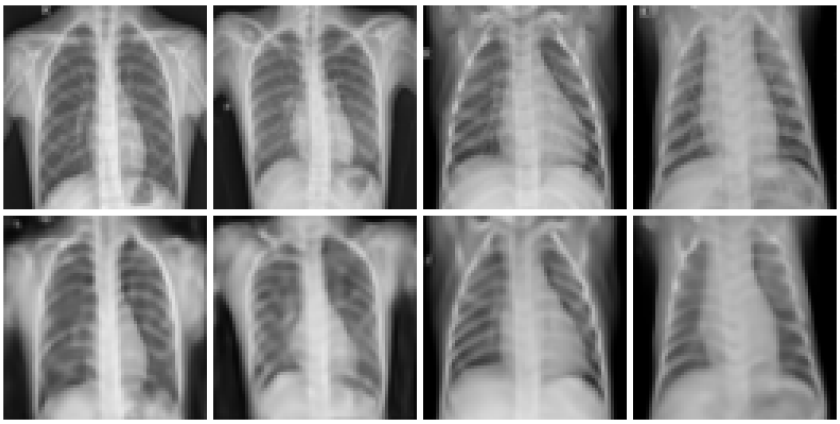


Figure 3: Four selected test images (top) and their reconstructions (bottom).

Overfitting appears around epochs 75–100: holdout reconstruction error rises while training error falls and KL stays relatively stable. Later checkpoints generated clearer ribs. Three sampling seeds gave holdout FIDs of 38.1 ± 1.1 at epoch 250 and 36.1 ± 1.4 at epoch 300 (mean and sample SD). Epoch 250 missed the predefined 1.5-FID tolerance for preferring an earlier checkpoint, so I kept epoch 300.

A nearest-neighbor check found no exact training-image matches among 5,000 samples; 1.54% fell below the holdout fifth-percentile distance. Pixel distance alone cannot rule out memorization.